

BEC303 Network Analysis — Syllabus mapping

EC branch, semester 3, 2025 scheme. Prescribed textbook: W. H. Hayt, J. E. Kemmerly and S. M. Durbin, *Engineering Circuit Analysis*, McGraw-Hill / Tata McGraw-Hill, 8th edition, 2012.

Course outcomes

CO	Statement
CO1	Analyze electronic networks using mesh and nodal analysis techniques for DC and AC circuits with dependent and independent sources.
CO2	Evaluate electronic circuit responses (currents, voltages, and power) using network theorems.
CO3	Analyze RL, RC, and RLC circuits under transient conditions using initial and final condition concepts.
CO4	Analyze linear electronic circuits using Laplace transform techniques.
CO5	Evaluate the performance of two-port networks by determining parameters and impedance characteristics.

Module plan

Module	Title	Hours	Chapters	Units	Deck slides	COs
1	Basic Concepts — Mesh and Node Analysis	15	Hayt 8e Ch. 1 (1.1), Ch. 2 (2.3), Ch. 3 (3.2–3.3), Ch. 4 (4.1–4.4), Ch. 5 (5.2, 5.5), Ch. 10 (10.1–10.6), pp. 2–397	16	56	CO1
2	Network Theorems for DC and AC Circuits	13	Hayt 8e Ch. 5 (5.1, 5.3, 5.4), Ch. 10 (10.7), pp. 123–154 and 397–406; reciprocity from syllabus TB2 (Channa Venkatesh & Ganesh Rao) §4.9	16	56	CO2
3	Transient Behavior and Initial Conditions	14	Hayt 8e Ch. 7 (7.1, 7.2, 7.7), Ch. 8 (8.1–8.9), Ch. 9 (9.1–9.7), Ch. 10 (10.1–10.7), pp. 217–247, 261–361 and 371–406	16	56	CO3
4	Laplace Transformation and Applications	14	Hayt 8e Ch. 14 (14.3–14.7), Ch. 15 (15.1–15.8), pp. 540–564 and 571–610; waveform and network synthesis from syllabus TB2 §6.8	16	56	CO4
5	Two-Port Networks	14	Hayt 8e Ch. 17 (17.1–17.6), pp. 687–720	16	56	CO5

Module 1 — Basic Concepts — Mesh and Node Analysis

Driving question: Given any linear network, how do I write the smallest set of equations that fully determines it — in DC and in AC?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	What circuit analysis actually is	Ch. 1	2-4	5, 6, 7	CO1	L2
2	Sources of electrical energy: independent and dependent	Ch. 2	17-22	8, 9, 10	CO1	L2
3	Kirchhoff's Current Law	Ch. 3	40-42	11, 12, 13	CO1	L2
4	Kirchhoff's Voltage Law	Ch. 3	42-46	14, 15, 16	CO1	L2
5	Nodal analysis: the standard procedure	Ch. 4	80-89	17, 18, 19	CO1	L3
6	The supernode	Ch. 4	89-92	20, 21, 22	CO1	L3
7	Mesh analysis: the standard procedure	Ch. 4	92-98	23, 24, 25	CO1	L3
8	The supermesh	Ch. 4	98-101	26, 27, 28	CO1	L3
9	Nodal and mesh analysis with dependent sources	Ch. 4	80-101	29, 30, 31	CO1	L4
10	Source transformations	Ch. 5	133-141	32, 33, 34	CO1	L3
11	Delta-wye conversion	Ch. 5	154-157	35, 36, 37	CO1	L3
12	Characteristics of sinusoids	Ch. 10	371-374	38, 39, 40	CO1	L2
13	From forced response to the complex forcing function	Ch. 10	374-383	41, 42, 43	CO1	L3
14	The phasor	Ch. 10	383-389	44, 45, 46	CO1	L3
15	Impedance and admittance	Ch. 10	389-394	47, 48, 49	CO1	L3
16	Nodal and mesh analysis of AC networks	Ch. 10	394-397	50, 51, 52	CO1	L4

Verbatim syllabus lines

- Basic Concepts- Mesh and Node Analysis: Introduction
- Source of Electrical Energy
- Kirchhoff's Laws
- Source transformations
- General Network Transformations
- Mesh and Node analysis with linearly dependent and independent sources for DC and AC networks
- Text book 1: 1.1, 3.2, 3.3, 4.1-4.4, 5.2, 5.5, 10.1-10.6
- Number of Hours: 15

Provenance

Built from the syllabus-listed Hayt 8e sections. Emphasis is on why nodal and mesh analysis are systematic rather than clever, on the supernode/supermesh repairs that keep them systematic when a source blocks the standard step, and on phasors as the device that makes the identical machinery work for sinusoidal steady state. Delta-wye and source transformation are presented as structure-changing shortcuts used before, not instead of, the systematic method.

Module 2 — Network Theorems for DC and AC Circuits

Driving question: When I do not need every branch current, which theorem gets me the one answer I want with the least work — and when does the shortcut stop being valid?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	Linearity: the property every theorem rests on	Ch. 5	123-133	5, 6, 7	CO2	L2
2	The superposition theorem	Ch. 5	123-133	8, 9, 10	CO2	L3
3	Deactivating sources correctly	Ch. 5	123-133	11, 12, 13	CO2	L2
4	Superposition with sources at different frequencies	Ch. 5	123-133	14, 15, 16	CO2	L4
5	Why superposition fails for power	Ch. 5	123-133	17, 18, 19	CO2	L4
6	The reciprocity theorem	TB2	\$4.9	20, 21, 22	CO2	L2
7	Conditions for reciprocity	TB2	\$4.9	23, 24, 25	CO2	L3
8	Reciprocal and non-reciprocal circuits	TB2	\$4.9	26, 27, 28	CO2	L4
9	Thévenin's theorem	Ch. 5	141-152	29, 30, 31	CO2	L3
10	Finding R_{th} : three methods and when each applies	Ch. 5	141-152	32, 33, 34	CO2	L4
11	Norton's theorem and Thévenin–Norton duality	Ch. 5	141-152	35, 36, 37	CO2	L3
12	The maximum power transfer theorem	Ch. 5	152-154	38, 39, 40	CO2	L4
13	Case study: maximum power transfer in electronic systems design	Ch. 5	152-154	41, 42, 43	CO2	L5
14	Network theorems in the phasor domain	Ch. 10	397-406	44, 45, 46	CO2	L4
15	Choosing between theorems: a decision procedure	Ch. 5	123-154	47, 48, 49	CO2	L5
16	Verifying a theorem result	Ch. 5	141-152	50, 51, 52	CO2	L5

Verbatim syllabus lines

- Network Theorems for DC and AC circuits: Introduction
- Superposition Theorem

- Reciprocity Theorem
- Conditions for Reciprocity
- Examples of Reciprocal/Non-Reciprocal circuits
- Thevenin's Theorem
- Maximum Power transfer theorem
- Case Study-1: Maximum Power Transfer in Electronic Systems Design
- Text book 1: 5.1, 5.3, 5.4, 10.7; Textbook 2: 4.9
- Number of Hours: 13

Provenance

Built from the syllabus-listed Hayt 8e sections plus the syllabus-listed TB2 section for reciprocity. Emphasis throughout is on the validity boundary of each theorem rather than its mechanics: superposition is exact for voltage and current and wrong for power; Thévenin requires linearity inside the port but nothing of the load; reciprocity fails the moment a dependent source appears. The maximum power transfer case study is developed as an engineering trade-off — maximum power is not maximum efficiency.

Module 3 — Transient Behavior and Initial Conditions

Driving question: A switch closes. What does every voltage and current in the circuit do between that instant and settling — and how much of it can I write down without solving anything?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	The capacitor: v-i relation, energy and continuity	Ch. 7	217-225	5, 6, 7	CO3	L2
2	The inductor: v-i relation, energy and continuity	Ch. 7	225-235	8, 9, 10	CO3	L2
3	Representing elements under switching	Ch. 7	217-247	11, 12, 13	CO3	L3
4	Evaluating initial conditions at $t = 0^+$	Ch. 8	261-268	14, 15, 16	CO3	L3
5	Evaluating final conditions at $t = \infty$	Ch. 8	275-282	17, 18, 19	CO3	L3
6	The source-free RL circuit and the time constant	Ch. 8	261-268	20, 21, 22	CO3	L3
7	Properties of the exponential response	Ch. 8	268-272	23, 24, 25	CO3	L3
8	The source-free RC circuit	Ch. 8	272-275	26, 27, 28	CO3	L3
9	The unit-step function and switched sources	Ch. 8	282-286	29, 30, 31	CO3	L3

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
10	Driven RL circuits: natural and forced response	Ch. 8	286-295	32, 33, 34	CO3	L4
11	Driven RC circuits and the power-ON delay case study	Ch. 8	295-300	35, 36, 37	CO3	L5
12	Sequentially switched circuits	Ch. 8	300-306	38, 39, 40	CO3	L4
13	The source-free parallel RLC and damping classification	Ch. 9	321-334	41, 42, 43	CO3	L4
14	Critically damped and underdamped responses	Ch. 9	334-345	44, 45, 46	CO3	L4
15	The source-free series RLC and the complete second-order response	Ch. 9	345-359	47, 48, 49	CO3	L4
16	Transients under AC excitation and the lossless LC circuit	Ch. 9	359-361	50, 51, 52	CO3	L5

Verbatim syllabus lines

- Transient behavior and initial conditions: Behavior of circuit elements under switching condition and their Representation
- Evaluation of initial and final conditions in RL, RC and RLC circuits for AC and DC excitations
- Case Study: RC Circuit-Based Power-ON Delay in Electronic Indicator Systems
- Text book 1: 7.1, 7.2, 7.7, 8.1 - 8.9, 9.1-9.7, 10.1-10.7
- Number of Hours: 14

Provenance

Built from the syllabus-listed Hayt 8e sections. The organising idea is that the two continuity rules — inductor current and capacitor voltage cannot jump — plus the DC limits at $t = \infty$ give three of the four things needed for any first-order answer without solving a differential equation at all. Second-order circuits are then presented as the same programme with a damping classification in front of it. Sequentially switched circuits are included because they are where students discover that the final condition of one interval is the initial condition of the next.

Module 4 — Laplace Transformation and Applications

Driving question: Can I replace calculus with algebra — solve the whole transient, initial conditions included, without ever writing a differential equation?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	Definition of the Laplace transform	Ch. 14	540-543	5, 6, 7	CO4	L2

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
2	Transforms of simple time functions	Ch. 14	543-546	8, 9, 10	CO4	L2
3	Inverse transform by partial fractions	Ch. 14	546-553	11, 12, 13	CO4	L3
4	Basic theorems: linearity, shifting, differentiation and integration	Ch. 14	553-561	14, 15, 16	CO4	L3
5	Initial-value and final-value theorems	Ch. 14	561-564	17, 18, 19	CO4	L4
6	Z(s) and Y(s): impedance in the s-domain	Ch. 15	571-578	20, 21, 22	CO4	L3
7	Initial conditions as sources in the s-domain	Ch. 15	571-578	23, 24, 25	CO4	L4
8	Nodal and mesh analysis in the s-domain	Ch. 15	578-585	26, 27, 28	CO4	L4
9	Thévenin, Norton and superposition in the s-domain	Ch. 15	585-588	29, 30, 31	CO4	L4
10	Transfer functions, poles and zeros	Ch. 15	588-589	32, 33, 34	CO4	L4
11	Step, ramp and impulse responses	Ch. 14	543-546	35, 36, 37	CO4	L4
12	Waveform synthesis	TB2	6.8	38, 39, 40	CO4	L4
13	Convolution	Ch. 15	589-598	41, 42, 43	CO4	L5
14	The complex-frequency plane and the first-order filter pole case study	Ch. 15	598-602	44, 45, 46	CO4	L5
15	Natural response and the s-plane	Ch. 15	602-606	47, 48, 49	CO4	L5
16	Network synthesis: realising a specified H(s)	Ch. 15	606-610	50, 51, 52	CO4	L5

Verbatim syllabus lines

- Laplace Transformation & Applications: Solution of networks
- step, ramp and impulse responses
- Waveform Synthesis
- Transfer function of elementary passive circuits
- Network Synthesis
- Case Study: Concept of poles for first order passive filters and their use in computing time response characteristics and bandwidth
- Text book 1: 14.3 - 14.7, 15.1-15.8; Textbook 2: 6.8

- Number of Hours: 14

Provenance

Built from the syllabus-listed Hayt 8e sections plus the syllabus-listed TB2 section for waveform and network synthesis. The through-line is that the s-domain does in one pass what Module 3 did in several: the natural and forced responses, the initial conditions and the steady state all emerge from one algebraic solution. Pole locations are introduced early and used repeatedly, so that by the case study students can read settling time and bandwidth off a pole position without computing a response at all.

Module 5 — Two-Port Networks

Driving question: How do I describe a network I cannot open, using only measurements made at its two accessible ports — and then predict what it does with any source and any load?

Topic to chapter, pages and deck slides

#	Teaching unit	Chapter	Pages	Deck slides	CO	Bloom
1	One-port networks and driving-point impedance	Ch. 17	687-692	5, 6, 7	CO5	L2
2	What a two-port is and why four numbers suffice	Ch. 17	687-692	8, 9, 10	CO5	L2
3	Short-circuit admittance (y) parameters	Ch. 17	692-699	11, 12, 13	CO5	L3
4	Equivalent networks and the π model	Ch. 17	699-708	14, 15, 16	CO5	L4
5	Open-circuit impedance (z) parameters	Ch. 17	708-713	17, 18, 19	CO5	L3
6	The relationship between z and y, and when each fails	Ch. 17	692-713	20, 21, 22	CO5	L4
7	Hybrid (h) parameters	Ch. 17	713-716	23, 24, 25	CO5	L3
8	Why multiple representations are required	Ch. 17	713-716	26, 27, 28	CO5	L5
9	Transmission (ABCD) parameters	Ch. 17	716-720	29, 30, 31	CO5	L4
10	Cascading two-ports with ABCD	Ch. 17	716-720	32, 33, 34	CO5	L4
11	Two-port symmetry and reciprocity	Ch. 17	699-708	35, 36, 37	CO5	L4
12	Input impedance of a terminated two-port	Ch. 17	708-713	38, 39, 40	CO5	L4
13	Output impedance and the source-terminated two-port	Ch. 17	708-713	41, 42, 43	CO5	L4
14	Measuring two-port parameters in practice	Ch. 17	692-713	44, 45, 46	CO5	L5
15	Choosing and converting between parameter sets	Ch. 17	692-720	47, 48, 49	CO5	L5
16	Putting it together: a fully terminated two-port	Ch. 17	708-720	50, 51, 52	CO5	L5

Verbatim syllabus lines

- Two-Port Network: Introduction
- Characterisation of Linear Time-invariant Two-port Networks
- Open-Circuit Impedance Parameters
- Short-circuit Admittance Parameters

- Transmission Parameters
- Short introduction to Hybrid parameters (1 hour material)
- Why multiple representations are required
- Two-port Symmetry
- Input impedance in terms of Two-port Parameters
- Output Impedance
- Text book 1: 17.1 - 17.6
- Number of Hours: 14

Provenance

Built from the syllabus-listed Hayt 8e Chapter 17 sections. The organising argument is that a two-port is a black box characterised by four numbers, and that the six possible parameter sets exist because each one is undefined for some perfectly ordinary network — which is why "why multiple representations are required" is treated as a unit in its own right rather than a remark. Reciprocity and symmetry are carried forward from Module 2 and shown here as matrix conditions. The module closes on terminated input and output impedance, which is the link back to the maximum power transfer results of Module 2.

Note on textbook page numbers

All TB1 chapter and page references were taken from the supplied Engineering Circuit Analysis PDF's own table of contents; the file extracted to roughly 2.0 million characters of machine-readable text, so no TB1 page reference in this document is from memory.

TB2 was not supplied in this environment. The syllabus lists K. Channa Venkatesh and D. Ganesh Rao, *Network Analysis - A Simplified Approach*, Cengage India, 2nd edition, 2012 for Module 2 (reciprocity, §4.9) and Module 4 (waveform and network synthesis, §6.8). Units depending on it are aligned to the verbatim syllabus headings, cite the chapter numbers the syllabus itself gives, and claim no page numbers. Every affected slide says so in its [Sources] note.